

```
1 class Kaprekar_Number
2 {   int i;
3     public Kaprekar_Number ()
4     {
5         i=Integer.MIN_VALUE;}
6
7     private int count_digits(int n)
8     {
9         if(n<10)
10            return 1;
11        return 1+count_digits(n/10);
12    }
13
14    private void kaprekar(int n)
15    {
16        int sq_n = n * n;
17        int count_digits = count_digits(sq_n);
18
19        for (int r_digits=1; r_digits<count_digits; r_digits++)
20        {
21            int eq_parts = (int) Math.pow(10, r_digits);
22            if (eq_parts == n)
23                continue;
24            int sum = sq_n/eq_parts + sq_n % eq_parts;
25            if (sum == n)
26                System.out.println(n + "\t"+n*n+"\t"+sq_n/eq_parts+"+"+ sq_n %
27                eq_parts );
28        }
29    }
30
31    void work()
32    {
33        System.out.println("THE KAPREKAR NUMBERS WITHIN 1000 ARE\n
34        NUMBER\tSQUARE\tSPLIT");
35
36        for ( i=1; i<=1000; i++)
37            if(i==1)
38                System.out.println(i + "\t"+i*i+"\t"+i/10+"+"+ 1 % 10 );
39            else
40                kaprekar(i);
41    }
42
43    public static void main()
44    {
45        Kaprekar_Number ob=new Kaprekar_Number();
46        ob.work();
47    }
48 }
49
```